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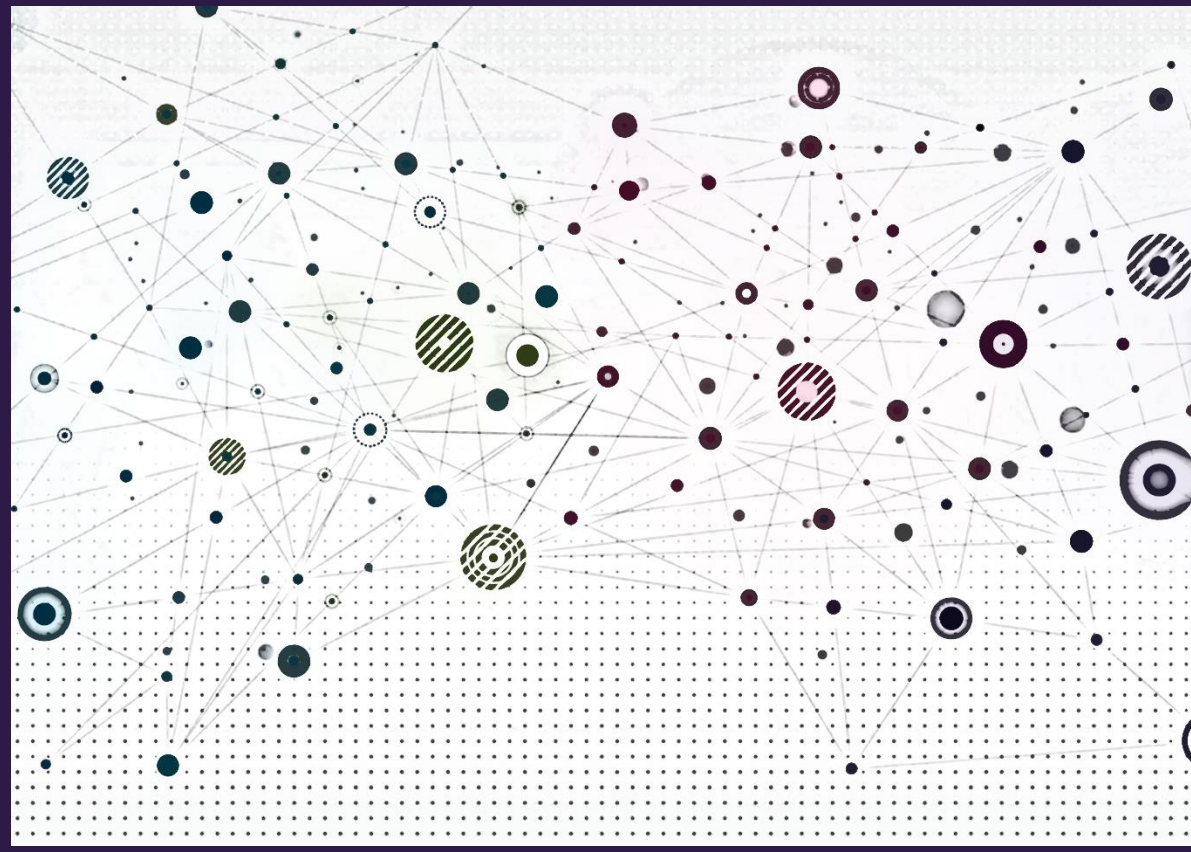
Knowledge graphs



Overview

- Recap
- Knowledge graphs
- Knowledge graph completion

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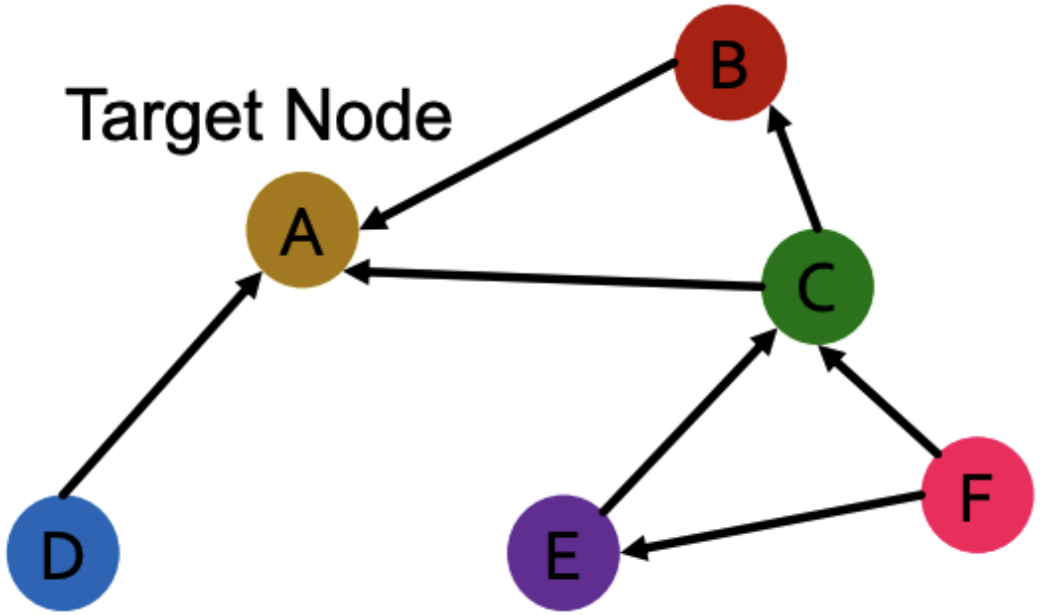
Recap

Heterogeneous graphs

- Multiple node and / or edge types yielding multiple *relation* types



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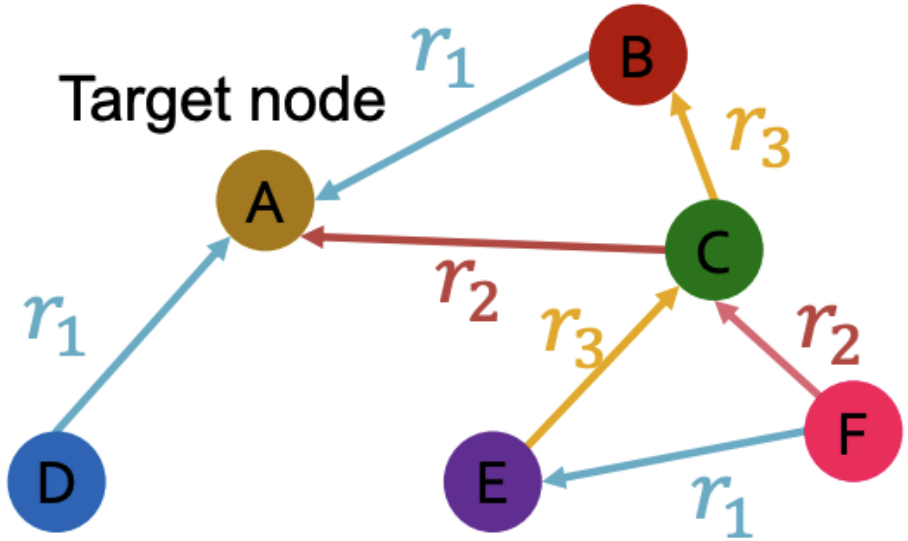
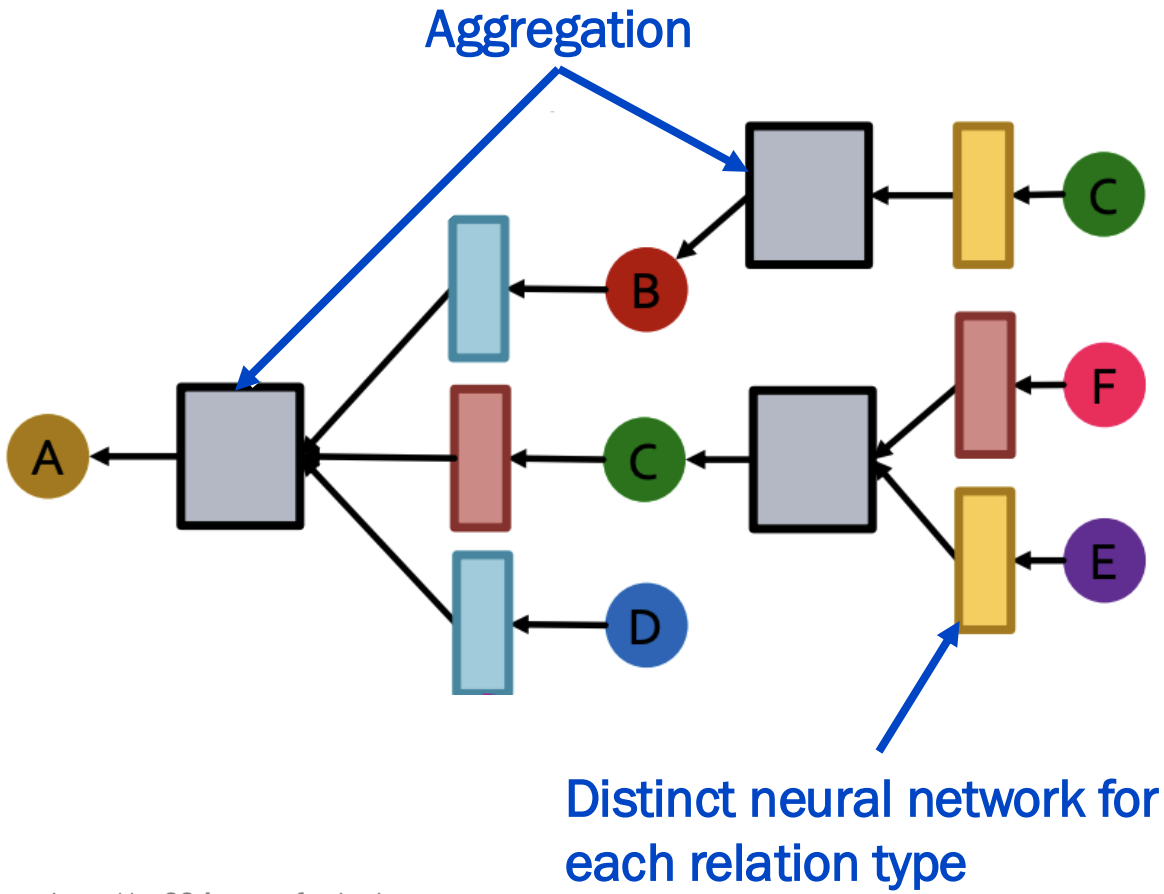
Relational graph convolutional networks (RGCN)

- Extend GCN to handle heterogeneous graphs with multiple relation types
- Different neural networks for different relation types





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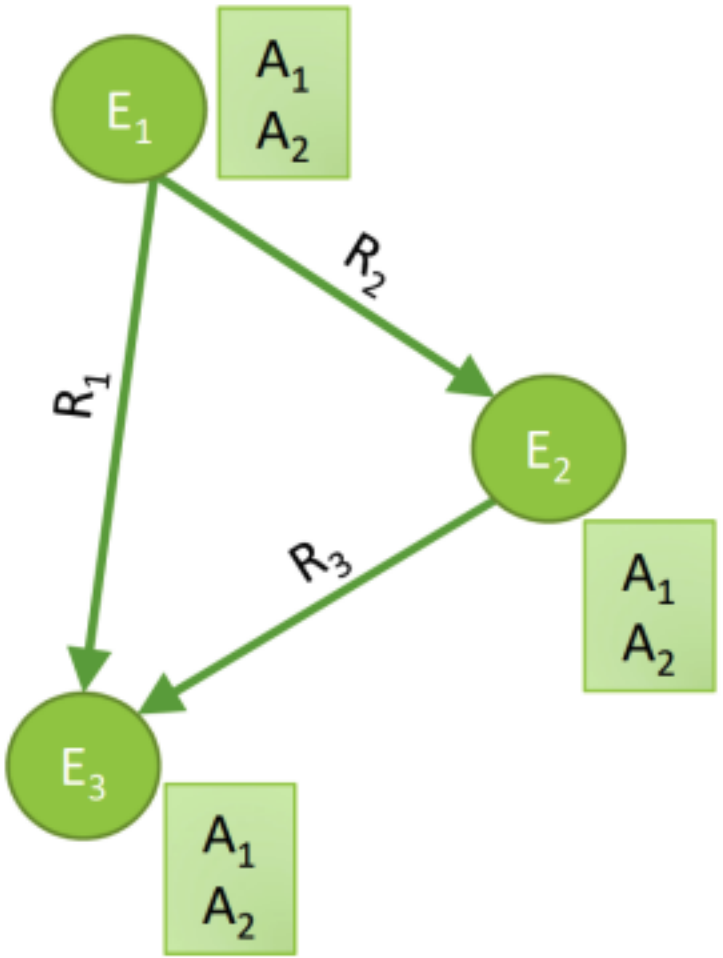
Knowledge graphs

Knowledge graphs

- Heterogeneous graphs that represent entities, types, and relationships
- Nodes are *entities*
- Nodes are labeled with their types
- Edges between nodes indicate specific relationships between entities, often directional



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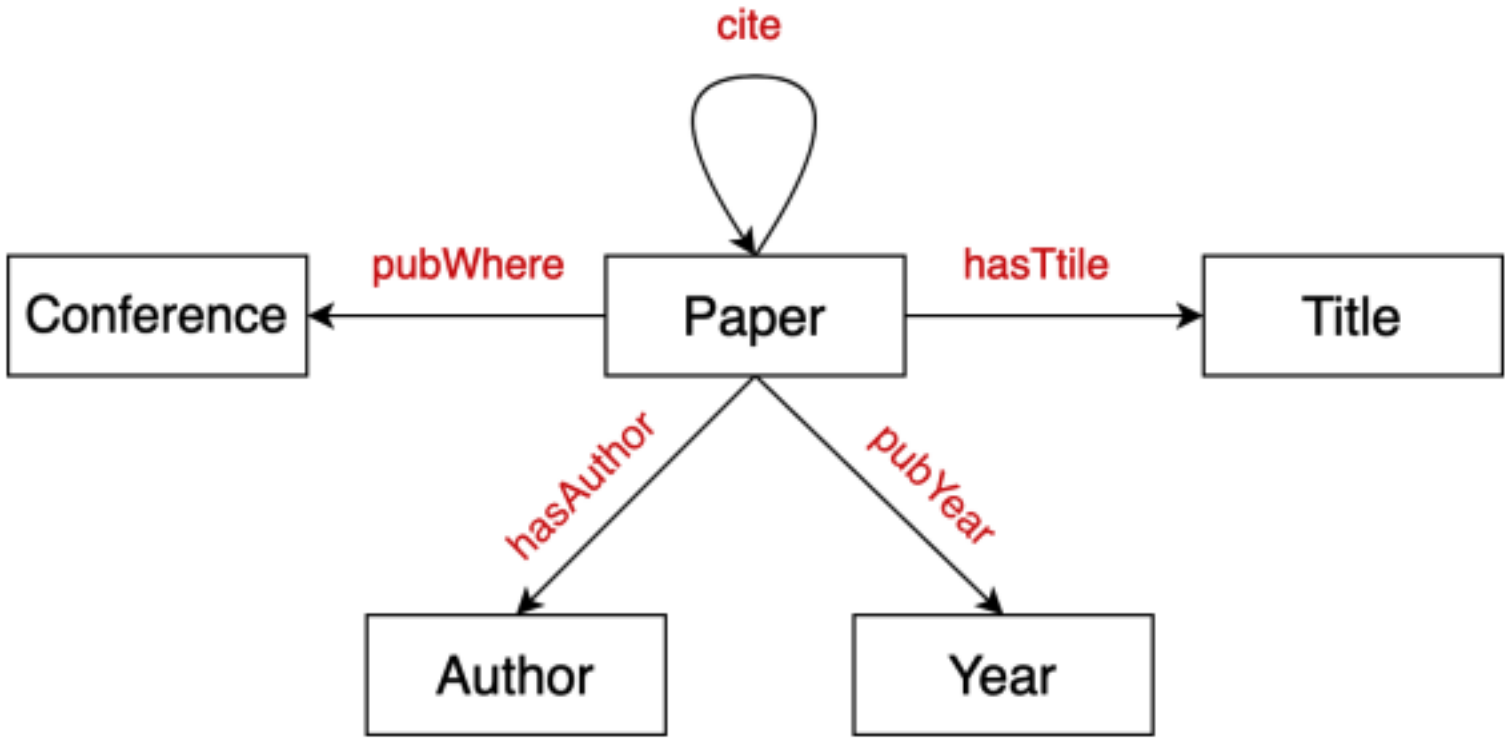
Bibliographic networks

- Node types: paper, title, author, conference, year
- Relation types: pubWhere, pubYear, hasTitle, hasAuthor, cite



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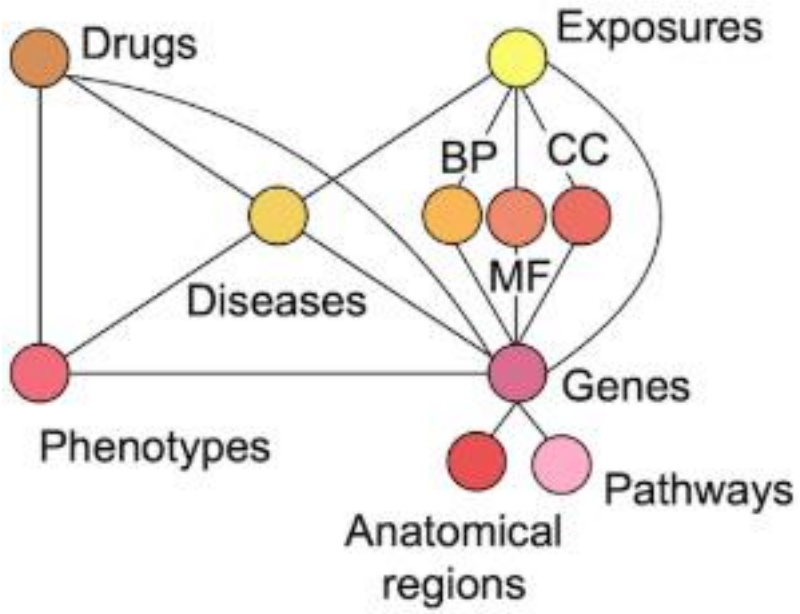
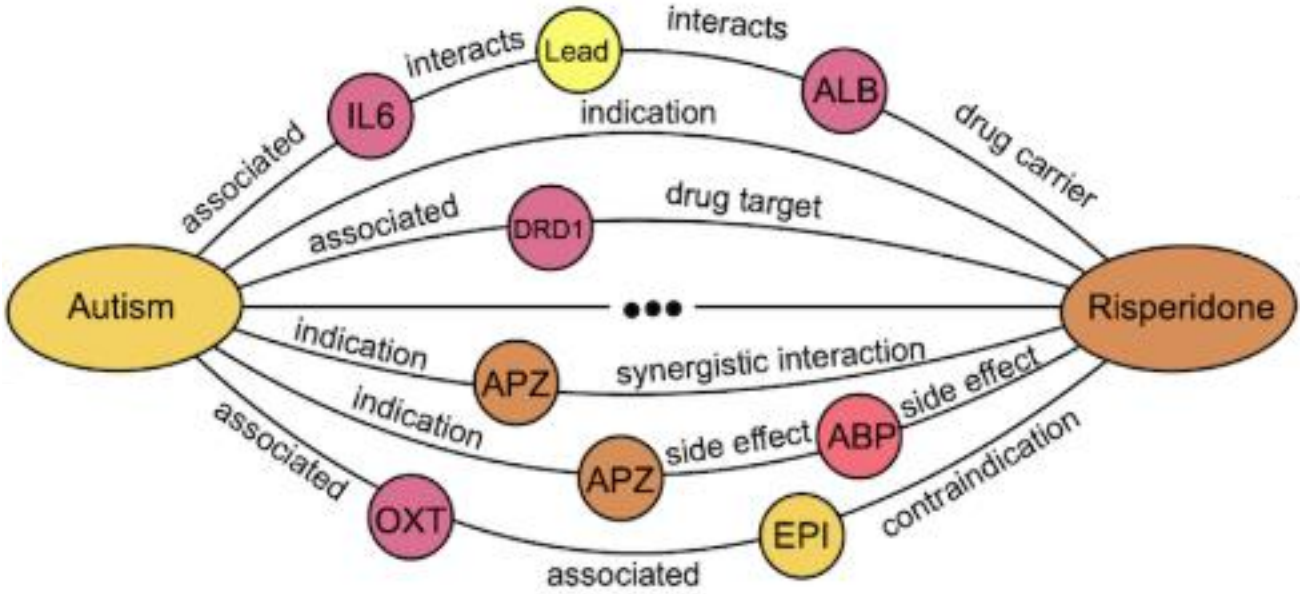
Biomedical knowledge graphs

- Node types: drugs, biological pathways, diseases, …
- Relation types: associated, interacts, …





More knowledge graphs

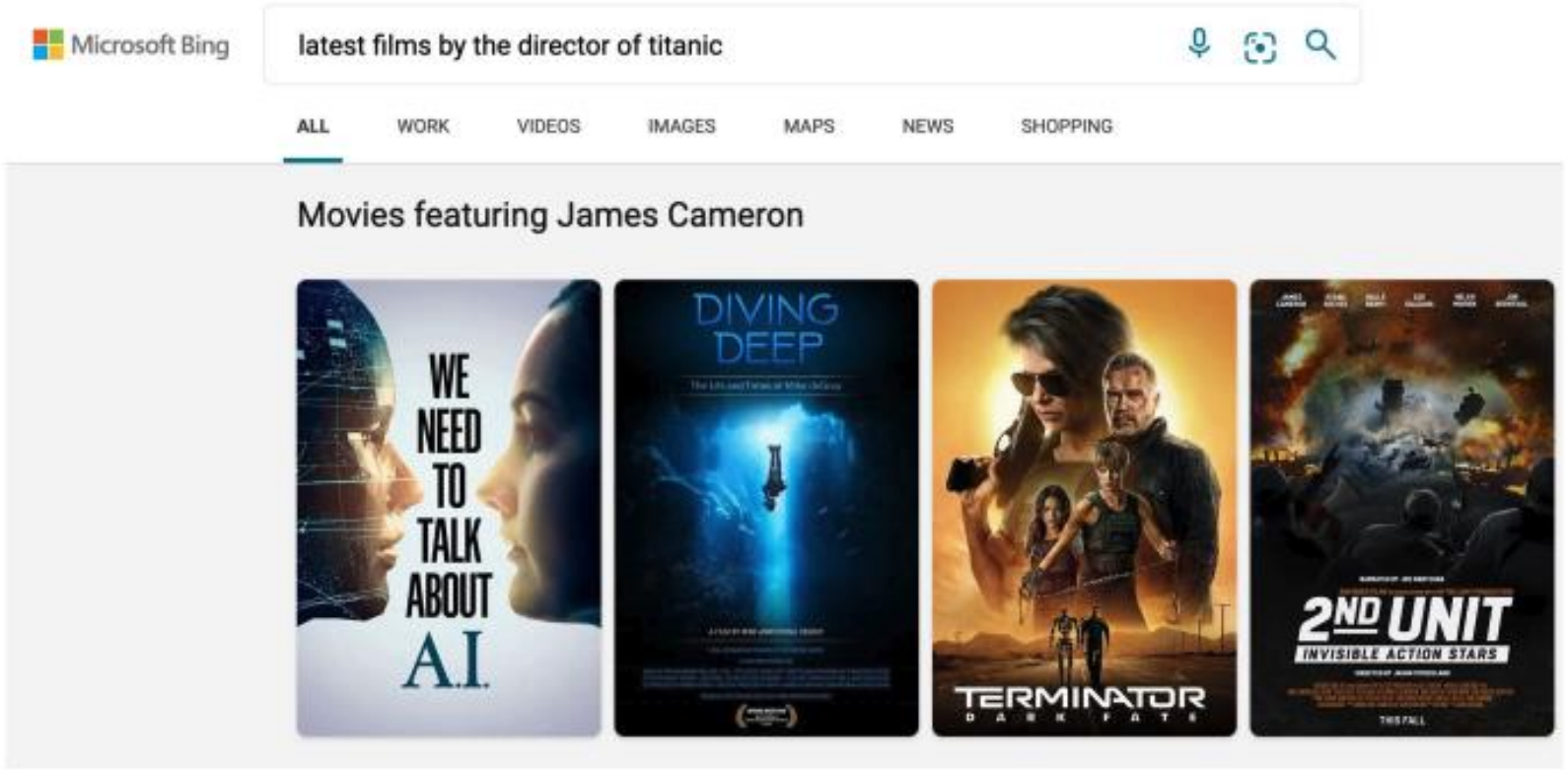
- Google knowledge graph - https://developers.google.com/knowledge-graph
- Amazon product graph
- Facebook graph API - https://developers.facebook.com/docs/graph-api/
- Microsoft graph - https://learn.microsoft.com/en-us/graph/overview
- LinkedIn knowledge graph

Publicly available knowledge graphs

- Wikidata - https://www.wikidata.org/wiki/Wikidata:Main_Page
- Dbpedia - https://www.dbpedia.org/resources/knowledge-graphs/
- YAGO - https://yago-knowledge.org/
- NELL - http://rtw.ml.cmu.edu/rtw/kbbrowser/

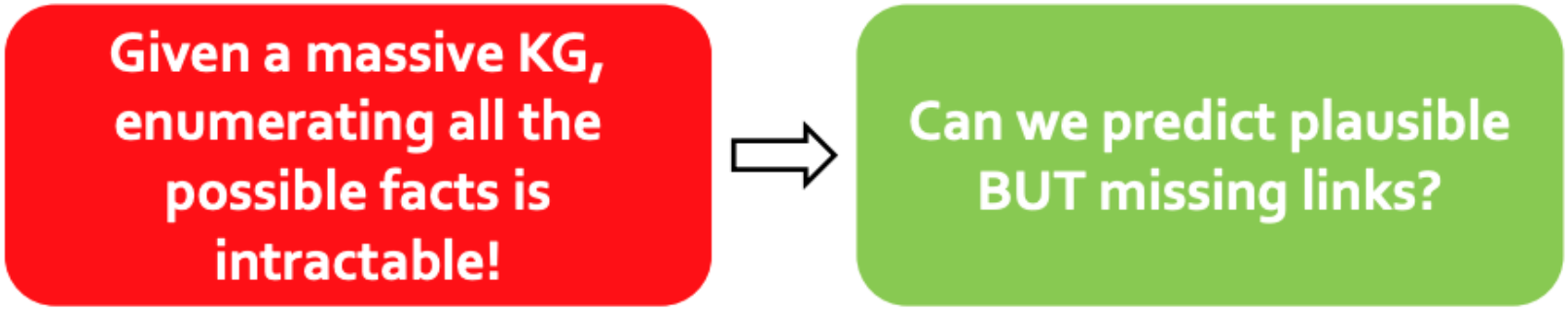
Applications of knowledge graphs

- Serving information
- Augmenting LLMs (we'll come back to this in a future lecture)



Common characteristics of modern knowledge graphs

- Massive: Millions (or Billions) of nodes and edges
  - Even scientific knowledge graphs have up to hundreds of thousands of nodes and edges
- Incomplete: Many true edges are missing (missing knowledge)



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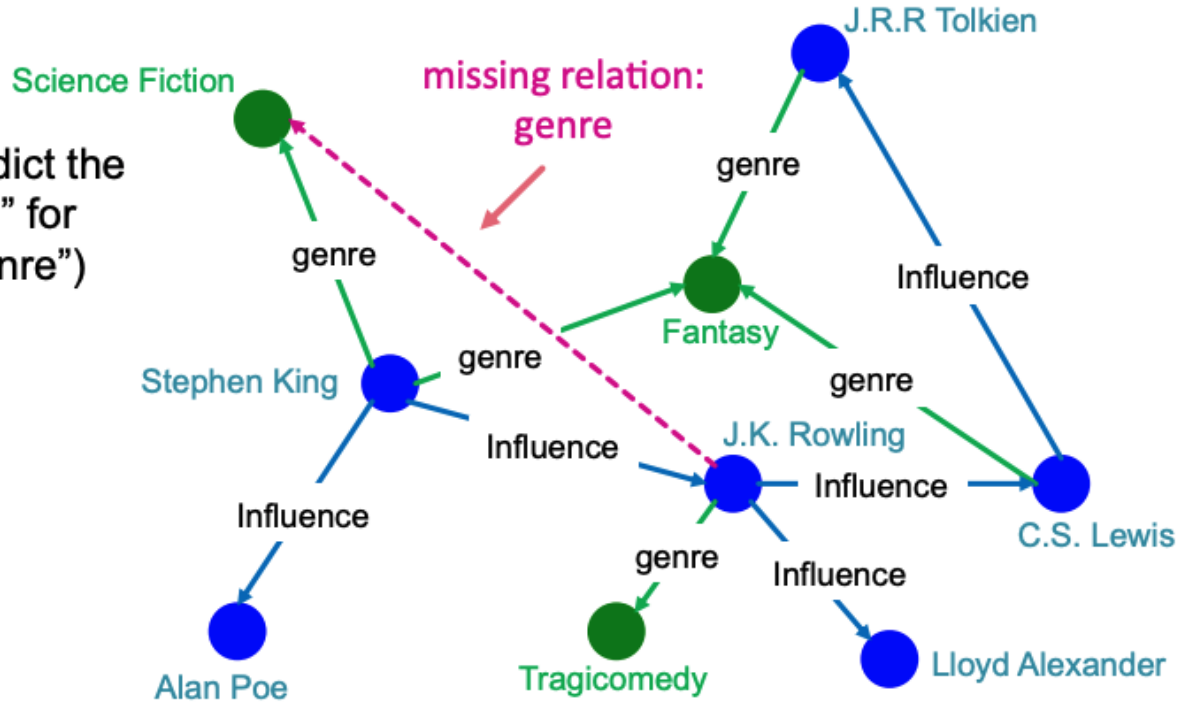
Knowledge graph completion

Knowledge graph (KG) completion task

- Given a large KG, can we ”complete” it (i.e., add missing edges)
- Task is typically formulated as predicting tails (or consequents)
  - Consider relations in a directed KG represented as a tuple (head, relation, tail) where head (h) has relation (r) with tail (t)
  - Given (head, relation) predict missing tails

Example task: predict the tail “Science Fiction” for (“J.K. Rowling”, “genre”)



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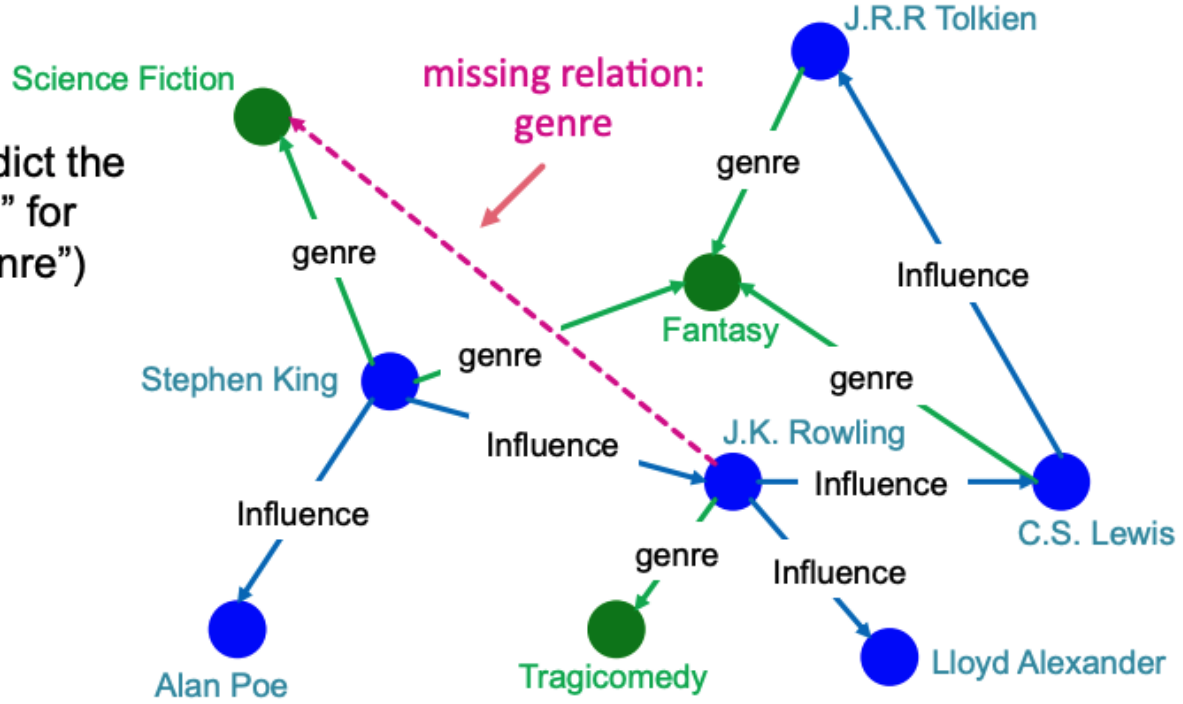
Knowledge graph (KG) completion task

- Model entities (nodes) and relations (edges) as embeddings in $\mathbb{R}^k$
- Given a triple *(h, r, t)*, the goal is to form an embedding of *(h, r)* that is close to the embedding of t
  - How to embed *(h, r)*?
  - How to define a score $f_r(h, t)$ that is high if *(h, r, t)* exists and low otherwise

Example task: predict the tail "Science Fiction" for ("J.K. Rowling", "genre")



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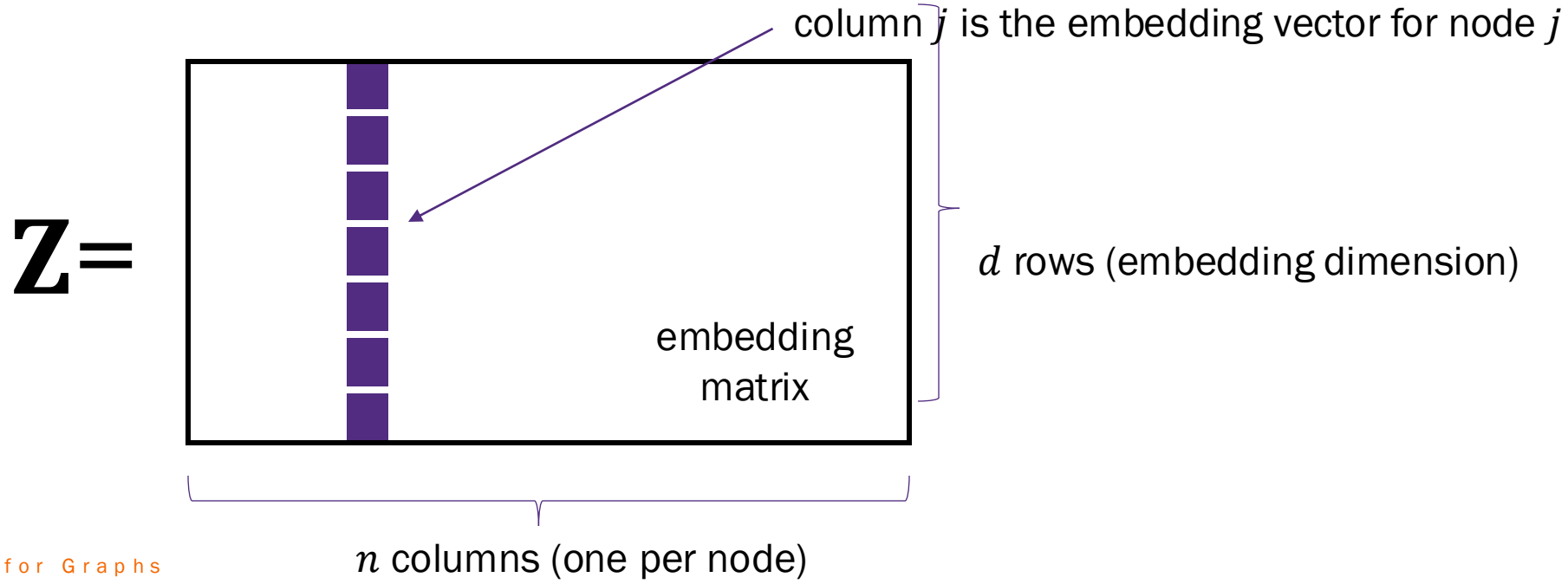
Transductive methods for KG completion

- "Standard" KG completion methods use shallow embedding methods for computational efficiency on large KGs
  - We'll focus on some of these that are based on different geometric intuitions
- Recently (<5 years) inductive GNN methods have been developed
  - See https://link.springer.com/article/10.1007/s00521-023-09286-2



Knowledge graph (KG) completion task

- We want our approaches to handle logical properties
  - Symmetric (Antisymmetric) $r(h, t) \Longrightarrow r(t, h)$, $r(h, t) \Longrightarrow !r(t,h)$
    - Example: Family (symmetric), Parent (antisymmetric)
    - NOTE: A relation cannot be symmetric and antisymmetric
  - Inverse: $r_1(h, t) \Longrightarrow r_2(t, h)$
    - Example: (Mentor, mentee)
  - Composition (Transitivity): $r_1(x, y) \wedge r_2(y, z) \Longrightarrow r_3(x, z)$
    - Example: parent (Joe, Linda) $\wedge$ mother (Linda, Steve) $\Longrightarrow$ grandParent(Joe, Steve) (the mother of my parent is my grandParent)
  - 1-to-N: $r(h, t_1)$, $r(h, t_2)$, ..., $r(h, t_n)$ are all True
    - Example: r is “enrolled in ML4G”

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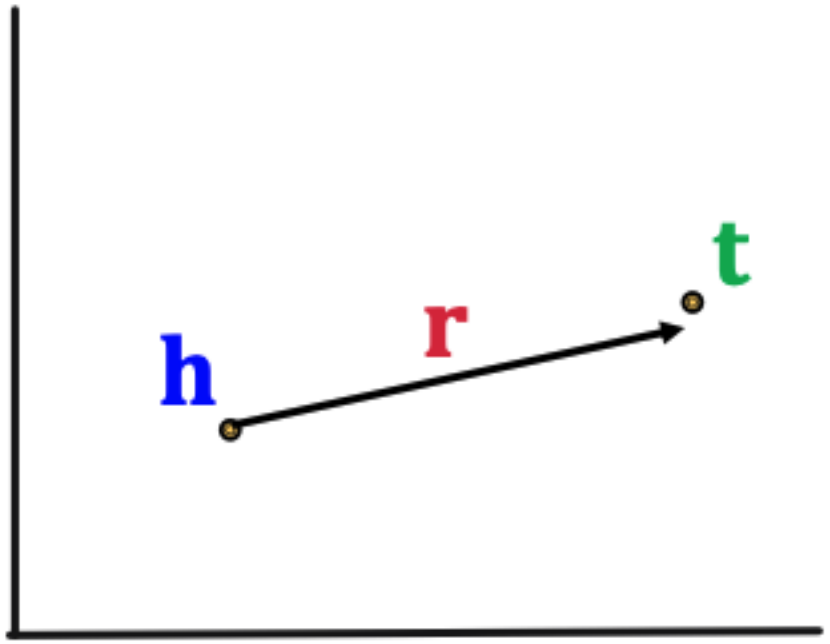
TransE method

Trans(lation) Embedding (TransE)

- Let $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$ be embeddings for the relation *(h, r, t)*
- **TransE:** $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ if the relation *(h, r, t)* exists, else $\mathbf{h} + \mathbf{r} \neq \mathbf{t}$
- Define the scoring function as $f_r(h, t) = -\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|$
- Geometric interpretation as vectors will help us examine the ability of **TransE** to model different logic relations



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TransE learning algorithm

Algorithm 1 Learning TransE

input Training set $S = \{(h,r,t)\}$, entities and rel. sets E and R, margin γ, embeddings dim. k.

initialize
$r \leftarrow \text{uniform}(-\frac{6}{\sqrt{k}}, \frac{6}{\sqrt{k}})$ for each $r \in R$
$r \leftarrow r / \|r\|$ for each $r \in R$
$\mathbf{e} \leftarrow \text{uniform}(-\frac{6}{\sqrt{k}}, \frac{6}{\sqrt{k}})$ for each entity $e \in E$

Initialize relations r and entities e uniformly, then normalize.
γ is the margin.

loop
  $\mathbf{e} \leftarrow \mathbf{e} / \|\mathbf{e}\|$ for each entity $e \in E$
  $S_{batch} \leftarrow$ sample(S, b) // sample a minibatch of size b
  $T_{batch} \leftarrow \emptyset$ // initialize the set of pairs of triplets
  for $(h,r,t) \in S_{batch}$ **do**
    $(h',r,t') \leftarrow$ sample$(S'_{(h,r,t)})$ // sample a corrupted triplet

Sample triplet (h',r,t) that does not appear in the KG.

    $T_{batch} \leftarrow T_{batch} \cup \{((h,r,t),(h',r,t'))\}$
  end for
  Update embeddings w.r.t.

$$\sum_{((h,r,t),(h',r,t')) \in T_{batch}} \nabla \left[\gamma + d(\boldsymbol{h}+\boldsymbol{r},\boldsymbol{t}) - d(\boldsymbol{h'}+\boldsymbol{r},\boldsymbol{t'})\right]_+$$

positive sample / negative sample

d represents distance (negative of score)

end loop

Contrastive loss: Favors lower distance (or higher score) for valid triplets, high distance (or lower score) for corrupted ones

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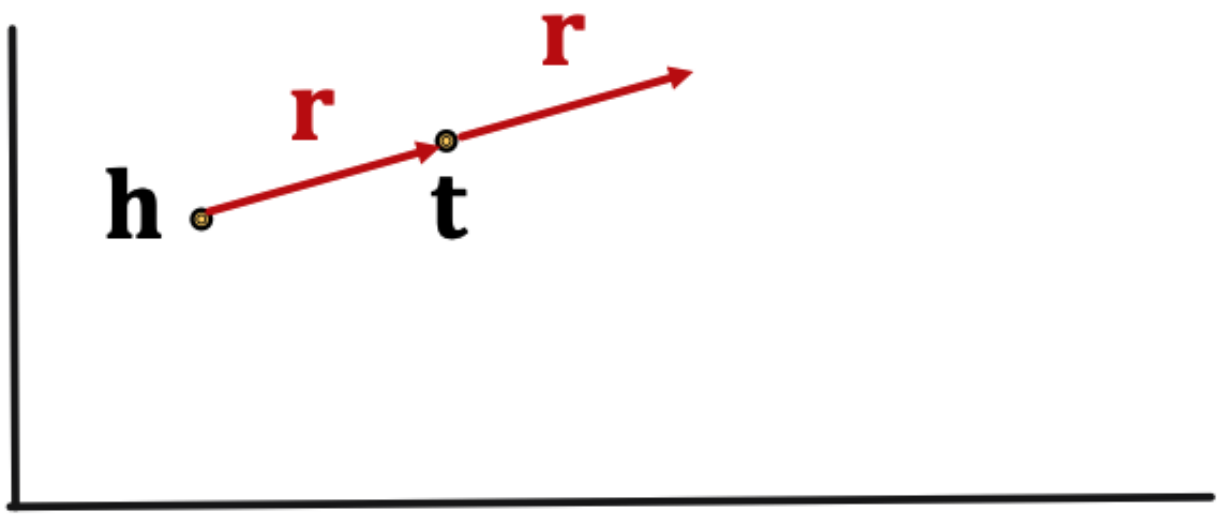
TransE antisymmetric relations

- r(h, t) $\Longrightarrow$!r(t, h)
- Example: parent(Joe, Linda) then !parent(Linda, Joe)
- TransE can model antisymmetric relations
  - $\mathbf{h} + \mathbf{r} = \mathbf{t}$ (by design), but $\mathbf{t} + \mathbf{r} \neq \mathbf{h}$



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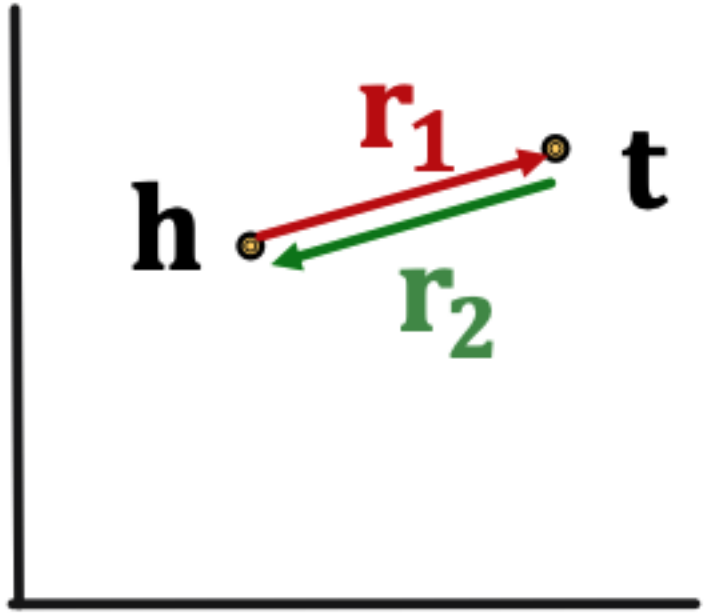
TransE inverse relations

- $r_1(h, t) \Longrightarrow r_2(t, h)$
- Example: (Mentor, mentee)
- TransE can model inverse relations
  - $\mathbf{h} + \boldsymbol{r_1} = \mathbf{t}$ and $\mathbf{t} + \boldsymbol{r_2} = \mathbf{h}$ by setting $\boldsymbol{r_1} = -\boldsymbol{r_2}$



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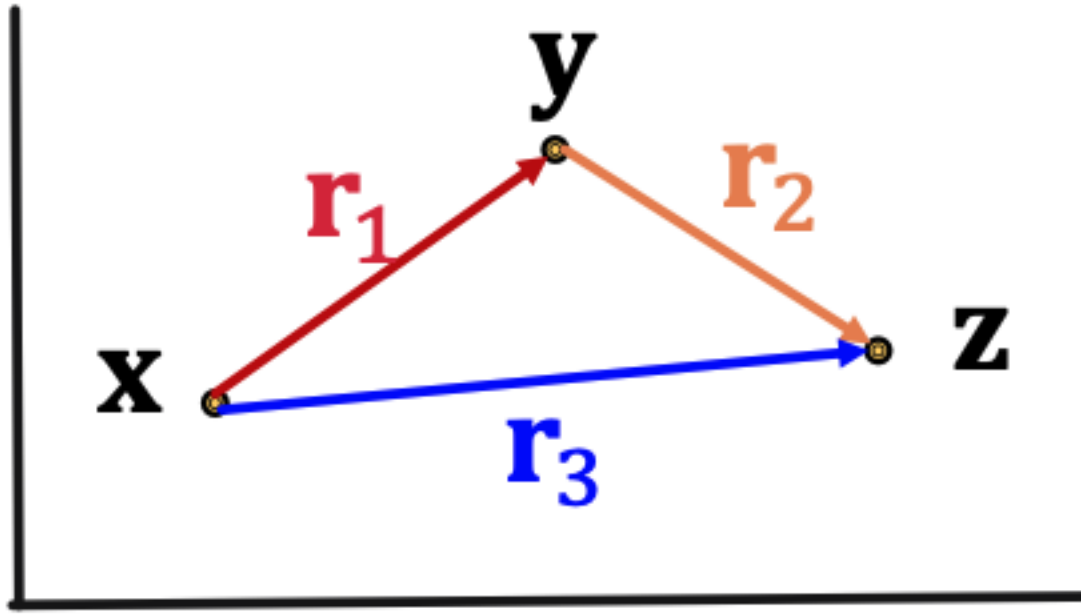
TransE composition relations

- $r_1(x, y) \wedge r_2(y, z) \Longrightarrow r_3(x, z)$
- Example: parent (Joe, Linda) Λ mother (Linda, Steve) ⟹ grandParent(Joe, Steve) (the father of my parent is my grandfather)
- TransE can model composition relations by setting $\boldsymbol{r_3 = r_1 + r_2}$



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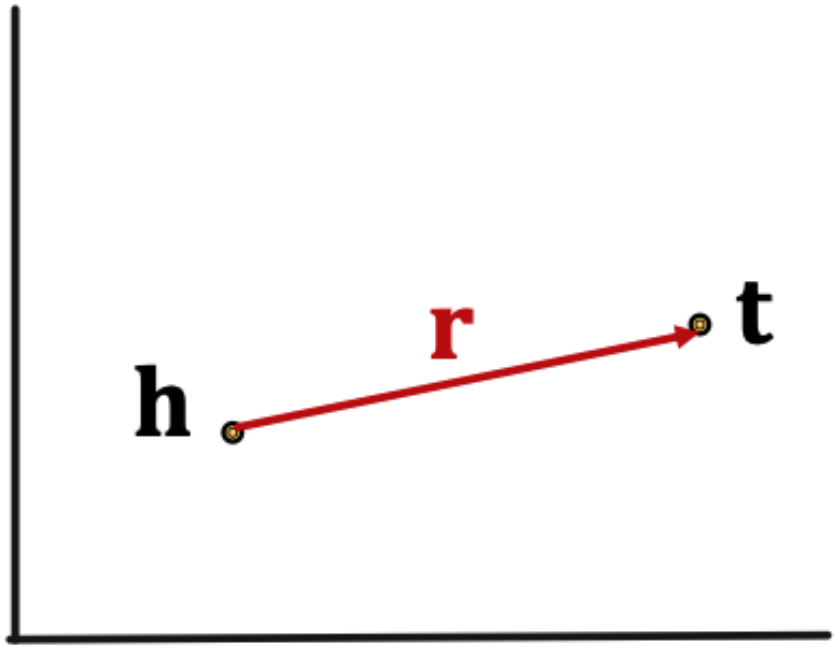
TransE symmetric relations

- $r(h, t) \Longrightarrow r(t, h)$
- Example: family
- TransE cannot model symmetric relations
  - Requires $\mathbf{h} + \mathbf{r} = \mathbf{t}$ and $\mathbf{t} + \mathbf{r} = \mathbf{h}$ which can only hold if $\mathbf{r} = \mathbf{0}$ and $\mathbf{h} = \mathbf{t}$



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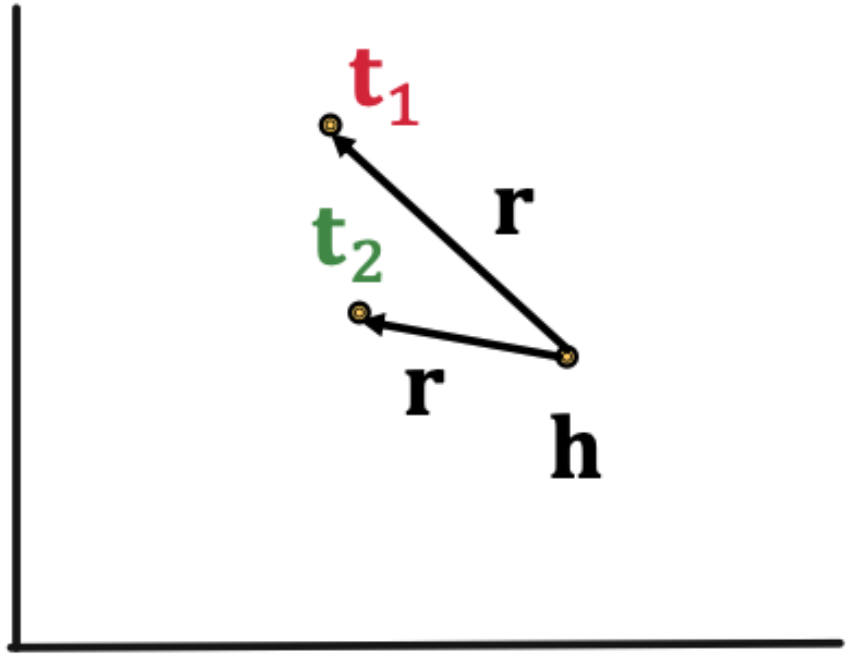
TransE 1-to-N relations

- Consider (h, r, t_1) and (h, r, t_2) both exist
- Example: t_1 and t_2 are enrolled in ML4G
- TransE cannot model both relations assuming $t_1 \neq t_2$
  - As t_1 and t_2 are different entities they should have different embeddings



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KG Completion Models

So far, we have

Model	Score	Embedding	Sym.	Antisym	Inv	Compos	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{r},\mathbf{t}\in\mathbb{R}^k$	✗	✓	✓	✓	✗

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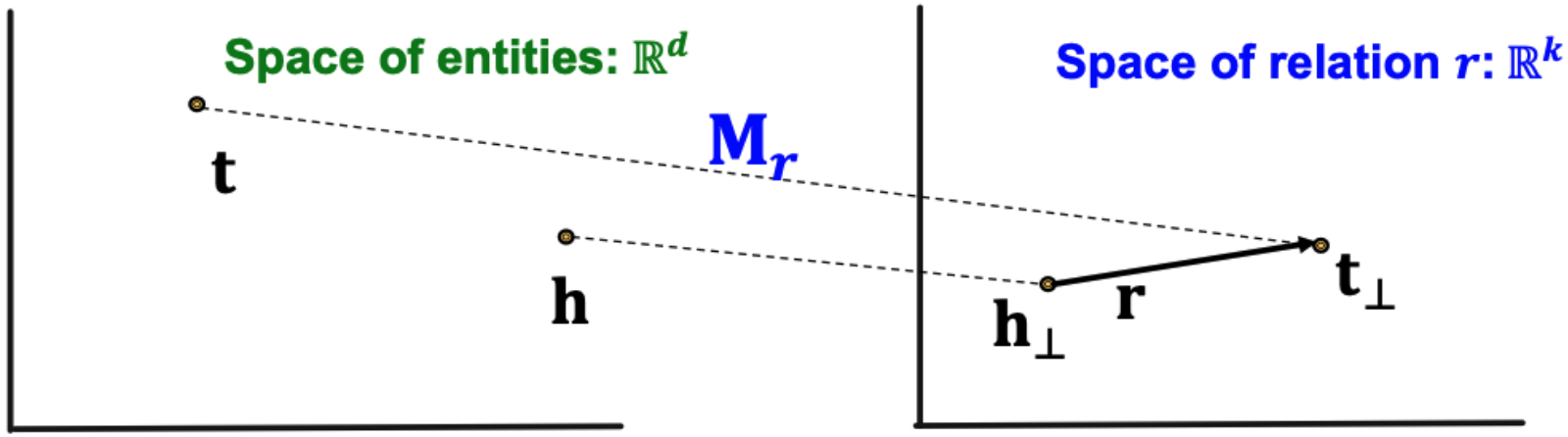
TransR method

TransR

- **TransE** conceptually models the translation of any relation in the same latent space
- **TransR** models entities as vectors in an entity space $\mathbb{R}^d$, and models each relation in its own relation space, $r \in \mathbb{R}^k$, using a projection matrix $M_r \in \mathbb{R}^{k \times d}$
- For any pair of relations r_1 and r_2, we can have $\mathbf{M}_{r_1} \neq \mathbf{M}_{r_2}$
  - We'll see that this gives us more flexibility than TransE

TransR

- **TransR** models entities as vectors in an entity space $\mathbb{R}^d$, and models each relation in its own relation space, $r \in \mathbb{R}^k$, using a projection matrix $M_r \in \mathbb{R}^{k \times d}$
- For a given $\mathbf{h}, \mathbf{r}, \mathbf{t}$, we set $\mathbf{h}_\perp = \mathbf{M}_r\mathbf{h}$ and $\mathbf{t}_\perp = \mathbf{M}_r\mathbf{t}$ to project $\mathbf{h}$ and $\mathbf{t}$ the entity space $\mathbb{R}^d$ to the relation space $\mathbb{R}^k$
- Score function: $f_r(h,t) = -\|\mathbf{h}_\perp + \mathbf{r} - \mathbf{t}_\perp\|$



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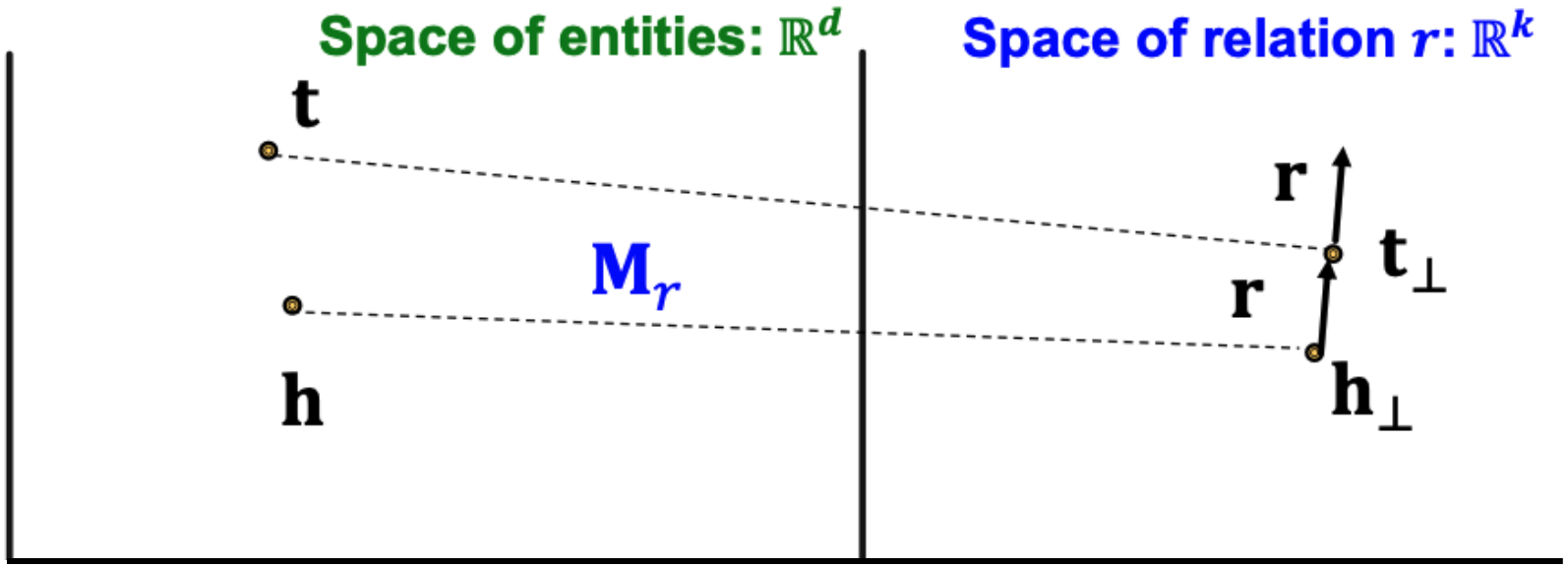
TransR antisymmetric relations

- r(h, t) $\Longrightarrow$!r(h,t)
- Example: parent(Joe, Linda) then !parent(Linda, Joe)
- TransR can model antisymmetric relations
  - $\mathbf{r} \neq \mathbf{0}$ and $\mathbf{M}_r\mathbf{h} + \mathbf{r} = \mathbf{M}_r\mathbf{t} = \mathbf{t}_\perp$ which implies $\mathbf{M}_r\mathbf{t} + \mathbf{r} \neq \mathbf{M}_r\mathbf{h} = \mathbf{h}_\perp$



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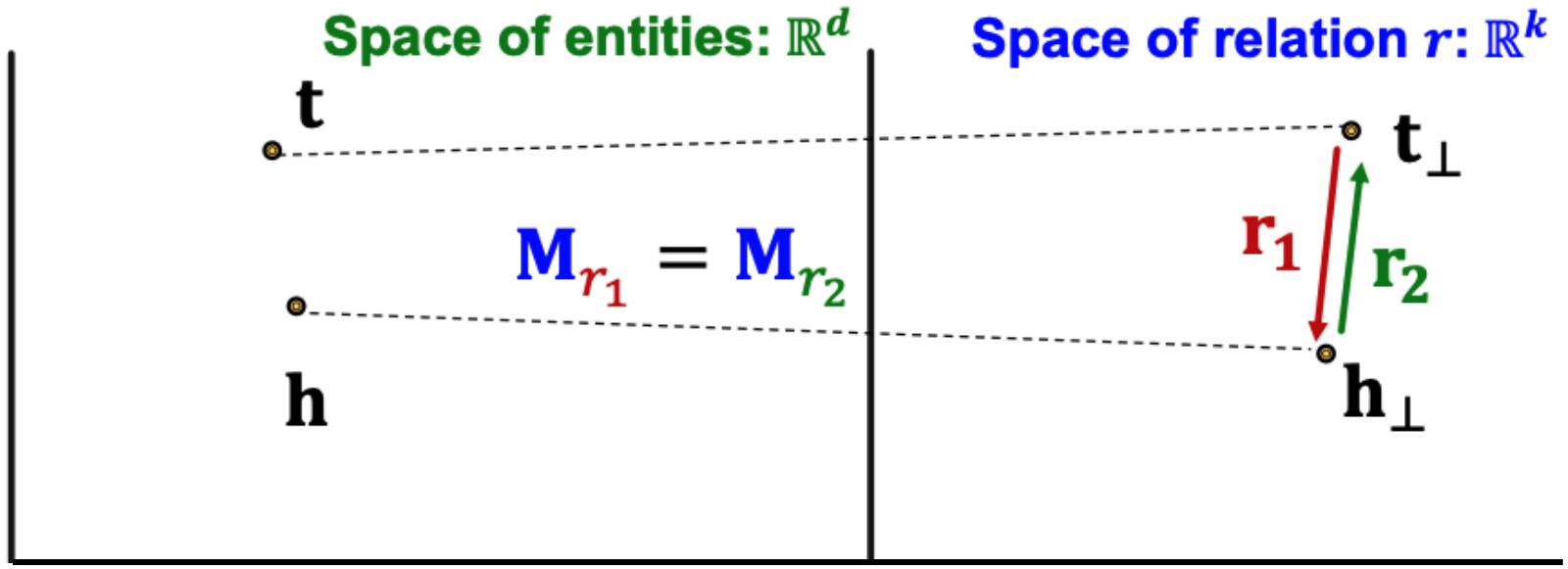
TransR inverse relations

- $r_1(h, t) \Longrightarrow r_2(t, h)$
- Example: (Mentor, mentee)
- TransR can model inverse relations

$$\boldsymbol{r_1 = -r_2}$$

Then $\mathbf{M}_{r_1}\mathbf{t} + \boldsymbol{r_1} = \mathbf{M}_{r_1}\mathbf{h}$ and $\mathbf{M}_{r_2}\mathbf{h} + \boldsymbol{r_2} = \mathbf{M}_{r_2}\mathbf{t}$



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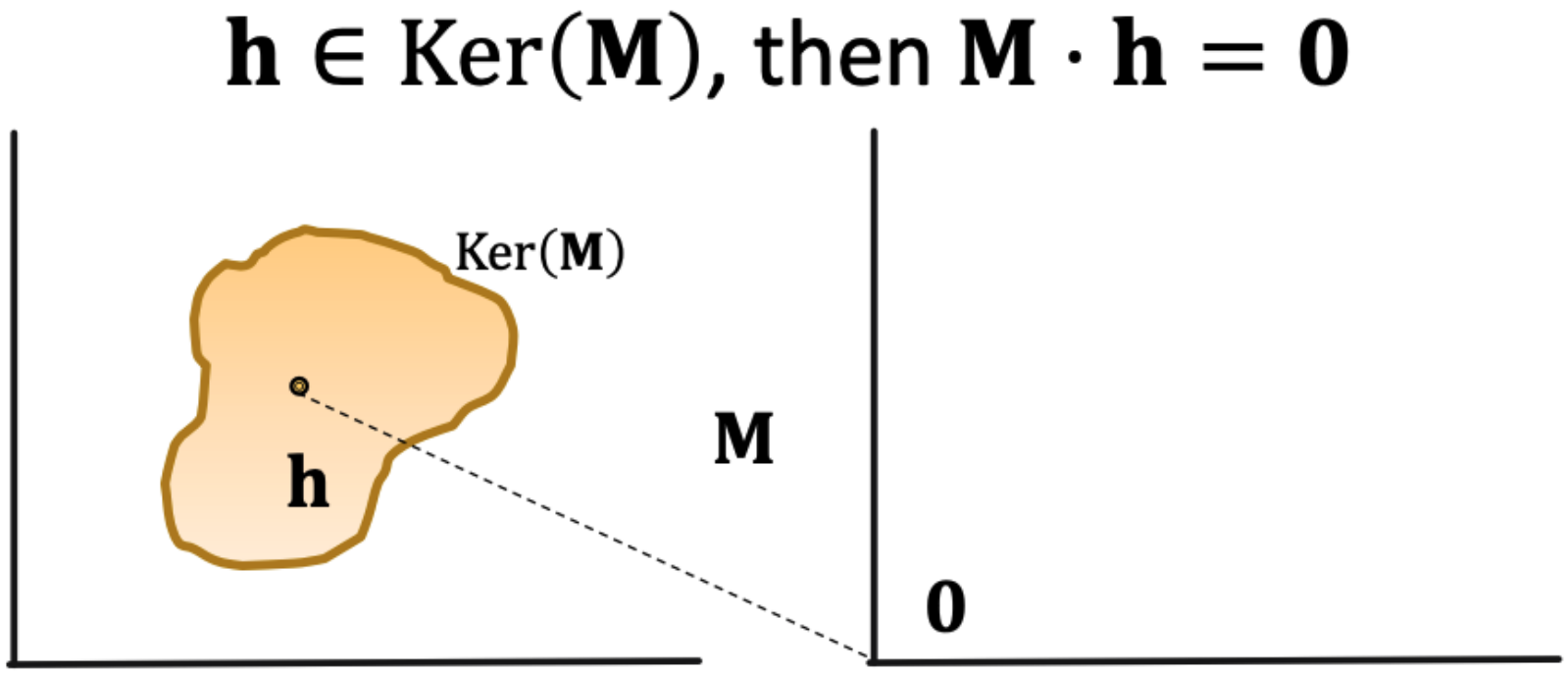
TransR and matrix null space

- Null space $N(\mathbf{M})$ (aka kernel space) of a matrix, $\mathbf{M}$, is the set of vectors that are mapped to the null (0) vector
- If $\mathbf{h} \in N(\mathbf{M})$ then $\mathbf{Mh} = \mathbf{0}$



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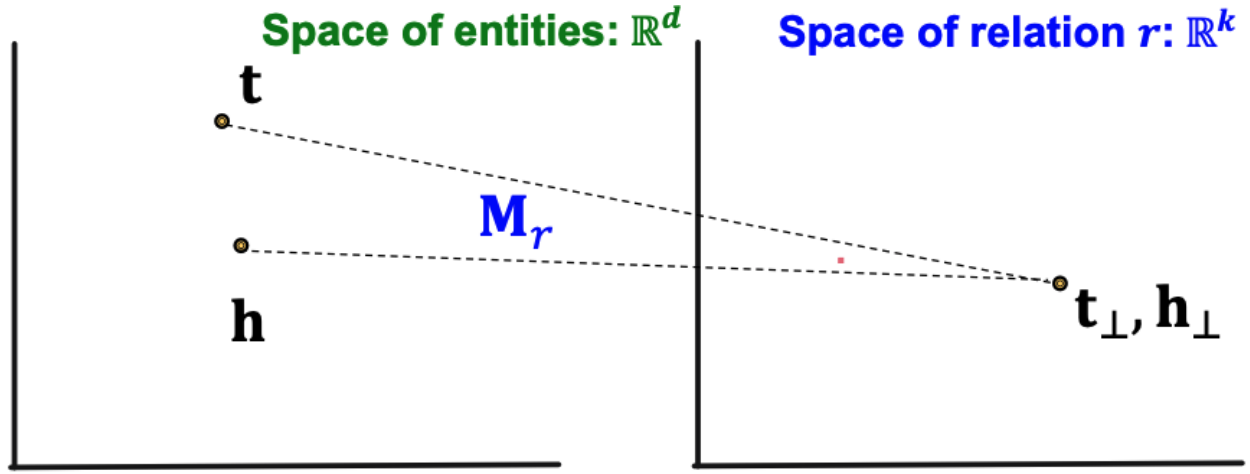
TransR symmetric relations

- r(h, t) $\Longrightarrow$ r(h, t)
- Example: family
- TransR can model symmetric relations
  - Set $\mathbf{r} = \mathbf{0}$ and $\mathbf{h}_\perp = \mathbf{M}_r\mathbf{h} = \mathbf{M}_r\mathbf{t} = \mathbf{t}_\perp$
- Requires $\mathbf{M}_r\mathbf{h} - \mathbf{M}_r\mathbf{t} = \mathbf{0}$ Implies $\mathbf{h} - \mathbf{t}$ is in the null space, $N(\mathbf{M}_r)$



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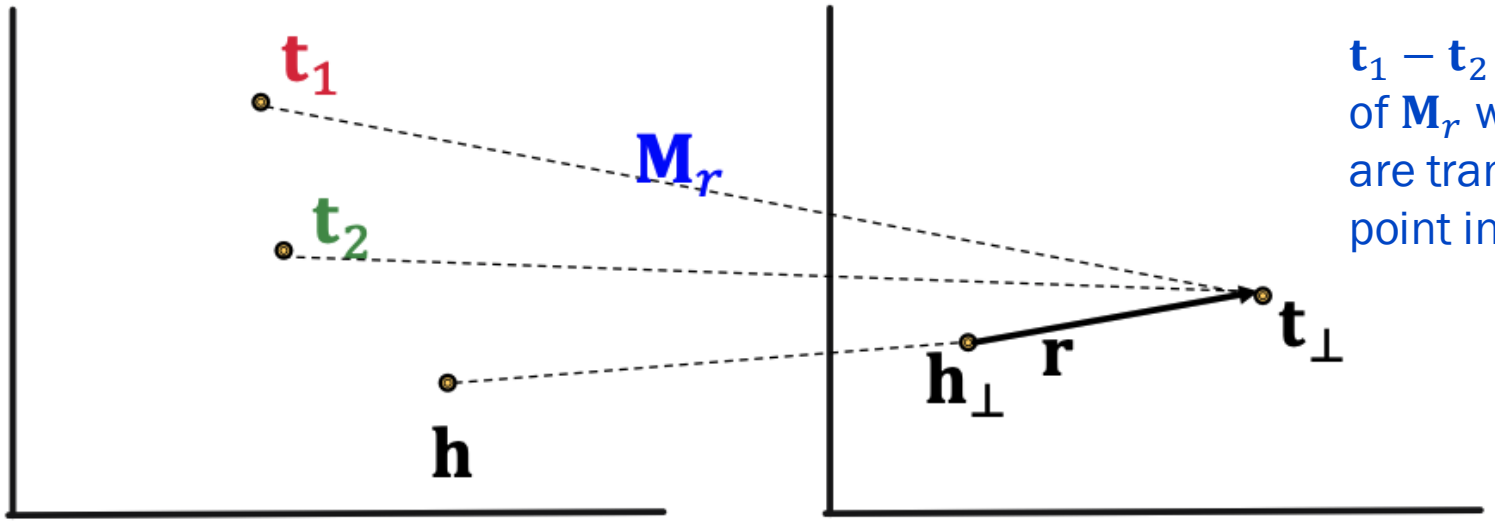
TransR 1-to-N relations

- Consider (h, r, t_1) and (h, r, t_2) both exist
- Example: t_1 and t_2 are enrolled in ML4G
- TransR can model both relations. Indeed, it can model any 1-to-N relation if we can find a matrix $\mathbf{M}_r$ such that for a given h, and for all entities t_i, t_j such that (h, r, t_i) and (h, r, t_j) exist, then $\mathbf{t}_i - \mathbf{t}_j$ is in the null space $N(\mathbf{M}_r)$ so that:

$$\mathbf{M}_r\mathbf{t}_i = \mathbf{M}_r\mathbf{t}_j$$



$\mathbf{t}_1 - \mathbf{t}_2$ is in the null space of $\mathbf{M}_r$ which means they are translated to the same point in the relation space

Image credit: Jure Leskovec, http://cs224w.stanford.edu

TransR composition relations

- $r_1(x, y) \wedge r_2(y, z) \Longrightarrow r_3(x, z)$
- Example: parent (Joe, Linda) $\wedge$ mother (Linda, Steve) $\Longrightarrow$ grandParent(Joe, Steve) (the mother of my parent is my grandParent)
- TransR can model composition relations
- Intuition: TransR models a triple with linear functions which are composable.
  - If $f(x)$ and $g(x)$ are linear, then the composition $f\big(g(x)\big)$ is also linear:

  Let $f(x) = ax + b$ and $g(x) = cx + d$ then $f\big(g(x)\big) = acx + (ad + b)$

- Incidentally, this is why non-linear activations are needed in deep learning, otherwise the multiple layers would just be equivalent to a linear function

TransR composition relations

- $r_1(x, y) \wedge r_2(y, z) \Longrightarrow r_3(x, z)$
- Assume $\mathbf{M}_{r_1}\mathbf{g}_1 = \mathbf{r}_1$ and $\mathbf{M}_{r_1}\mathbf{g}_2 = \mathbf{r}_2$

$\mathbf{g}_1$ is a vector subspace of the entity embedding space. $\mathbf{M}_{r_1}\mathbf{g}_1 = \mathbf{r}_1$ implies for any vector $\mathbf{g} \in \mathbf{g}_1$ that $\mathbf{g}_1 - \mathbf{r}_1 \in N(\mathbf{M}_{r_1})$. The same is true for $\mathbf{g}_2$.

  - If $r_1(x, y)$ exists $\Longrightarrow \mathbf{M}_{r_1}\mathbf{x} + \mathbf{r}_1 = \mathbf{M}_{r_1}\mathbf{y} \Longrightarrow \mathbf{M}_{r_1}(\mathbf{y} - \mathbf{x}) = \mathbf{r}_1$

$$\mathbf{y} - \mathbf{x} \in \mathbf{g}_1 + N(\mathbf{M}_{r_1}) \Longrightarrow \mathbf{y} \in \mathbf{x} + \mathbf{g}_1 + N(\mathbf{M}_{r_1})$$

  - If $r_2(y, z)$ exists $\Longrightarrow \mathbf{M}_{r_2}\mathbf{y} + \mathbf{r}_2 = \mathbf{M}_{r_2}\mathbf{z} \Longrightarrow \mathbf{M}_{r_2}(\mathbf{z} - \mathbf{y}) = \mathbf{r}_2$

$$\mathbf{z} - \mathbf{y} \in \mathbf{g}_2 + N(\mathbf{M}_{r_2}) \Longrightarrow \mathbf{z} \in \mathbf{y} + \mathbf{g}_2 + N(\mathbf{M}_{r_2})$$

  - Then, we have

$$\mathbf{z} \in \mathbf{x} + \mathbf{g}_1 + \mathbf{g}_2 + N(\mathbf{M}_{r_1}) + N(\mathbf{M}_{r_2})$$

TransR composition relations

- $r_1(x, y) \wedge r_2(y, z) \Longrightarrow r_3(x, z)$
- From the previoius slide:

$\mathbf{g}_1$ is a vector subspace of the entity embedding space. $\mathbf{M}_{r_1}\mathbf{g}_1 = \mathbf{r}_1$ implies for any vector $\mathbf{g} \in \mathbf{g}_1$ that $\mathbf{g}_1 - \mathbf{r}_1 \in N(\mathbf{M}_{r_1})$. The same is true for $\mathbf{g}_2$.

$$\mathbf{z} \in \mathbf{x} + \mathbf{g}_1 + \mathbf{g}_2 + N(\mathbf{M}_{r_1}) + N(\mathbf{M}_{r_2})$$

- We now construct $\mathbf{M}_{r_3}$ such that

$$N(\mathbf{M}_{r_3}) = N(\mathbf{M}_{r_1}) + N(\mathbf{M}_{r_2})$$

and set

$$\mathbf{r}_3 = \mathbf{M}_{r_3}(\mathbf{g}_1 + \mathbf{g}_2)$$

$\mathbf{g}_1 + \mathbf{g}_2 - \mathbf{r}_3$ is a vector subspace of the entity embedding space and all vectors in this subspace are in the null space $N(\mathbf{M}_{r_3})$

- Finally, we have

$$\mathbf{M}_{r_3}(\mathbf{z}) \in \mathbf{M}_{r_3}\mathbf{x} + \mathbf{M}_{r_3}(\mathbf{g}_1 + \mathbf{g}_2) + \mathbf{M}_{r_3}\left(N(\mathbf{M}_{r_1}) + N(\mathbf{M}_{r_2})\right)$$

necessarily a null vector

$$\mathbf{M}_{r_3}(\mathbf{z}) \in \mathbf{M}_{r_3}\mathbf{x} + \mathbf{M}_{r_3}(\mathbf{g}_1 + \mathbf{g}_2) = \mathbf{M}_{r_3}\mathbf{x} + \mathbf{r}_3$$

$$\mathbf{M}_{r_3}(\mathbf{z}) = \mathbf{M}_{r_3}(\mathbf{x}) + \mathbf{r}_3$$

KG Completion Models

Model	Score	Embedding	Sym.	Antisym	Inv	Compos	1-to-N
TransE	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|$	$\mathbf{h},\mathbf{r},\mathbf{t}\in\mathbb{R}^k$	✗	✓	✓	✓	✗
TransR	$-\|\|\mathbf{h}_\perp+r-\mathbf{t}_\perp\|\|$	$\mathbf{h},\mathbf{t}\in\mathbb{R}^d$ $\mathbf{r}\in\mathbb{R}^k$ $\mathbf{M}_r\in\mathbb{R}^{k\times d}$	✓	✓	✓	✓	✓

KG completion in practice

- There are numerous other methods for KG completion
- Different KGs may have drastically different relation patterns
- There is not a general method (yet) that works for all KGs
- Try TransE for a quick analysis
- Then use more complex (TransR, ComplEX, RotateE, …)